Electronics and Computer Technology

ECET 250E: Lab 06 First Order RC and RL Circuits

OBJECTIVE:

- Observe the transient response of series RC and RL circuits driven by a square wave, and explain the waveforms using the time constant τ.
- Measure τ for each circuit from oscilloscope waveforms and compare it with the value calculated from the component values (RC and L/R).
- Explain how the length of each half-period, compared with 5τ, decides whether the capacitor fully charges and discharges and whether the inductor current fully builds up and decays.

BACKGROUND:

Key ideas

A capacitor stores energy in its electric field, and an inductor stores energy in its magnetic field. Stored energy cannot change in zero time, so the **capacitor voltage** v_C and the **inductor current** i_L cannot jump. They can only change gradually. Every other quantity (i_C, v_L, v_R) can jump at the instant the source switches.

A circuit with resistors and one storage element is a *first-order circuit*. When the source switches, every voltage and current x(t) moves exponentially from its initial value x(0) to its final value x(∞):

$$x(t) = x(\infty) + \left[x(0) - x(\infty)\right] e^{-t/\tau} \qquad (1)$$

The time constant τ sets how fast this happens: $\tau = RC$ for an RC circuit and $\tau = L/R$ for an RL circuit, where R is the resistance seen by the capacitor or inductor. After one τ the response has covered 63.2 % of the change. After **5τ** less than 1 % of the change is left ($e^{-5} \approx 0.007$), so in practice the transient is over. At DC steady state a capacitor behaves as an *open circuit* and an inductor as a *short circuit*.

Series RC and RL circuits driven by a step

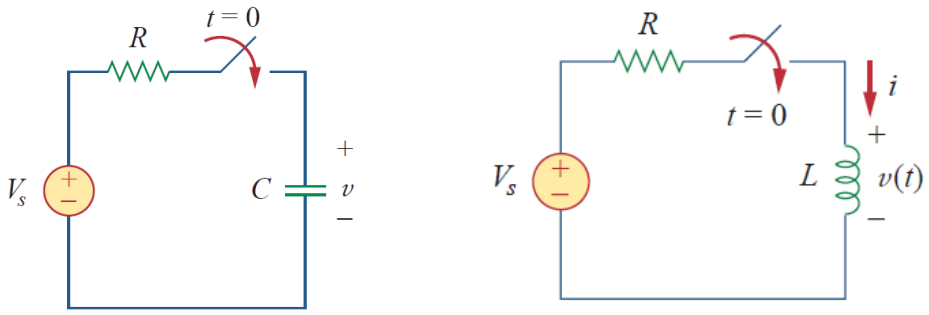


Figure 1: Series RC circuit (left) and series RL circuit (right) driven by a DC source V_s

Applying equation (1) to the circuits in Figure 1 gives the results in Table 1. "Charging" means the source steps from 0 to V_s with no stored energy. "Discharging" means the source drops to 0 V and the stored energy (initial V_0 or I_0) drains through the resistor; this is also called the *natural* or *source-free* response.

Table 1: Step response of series RC and RL circuits

	Series RC	**Series RL**
Time constant	$\tau = RC$	$\tau = L/R$
Cannot jump	capacitor voltage v_C	inductor current i_L
Charging (source 0 → Vs)	$v_C = V_s\left(1 - e^{-t/\tau}\right)$ $v_R = V_s e^{-t/\tau}$	$i_L = \frac{V_s}{R}\left(1 - e^{-t/\tau}\right)$ $v_R = V_s\left(1 - e^{-t/\tau}\right)$ $v_L = V_s e^{-t/\tau}$
Discharging (source → 0 V)	$v_C = V_0 e^{-t/\tau}$ $v_R = -V_0 e^{-t/\tau}$	$i_L = I_0 e^{-t/\tau}$ $v_R = R I_0 e^{-t/\tau}$ $v_L = -R I_0 e^{-t/\tau}$
DC steady state	C acts as an open circuit: i = 0, $v_C = V_s$	L acts as a short circuit: $v_L = 0$, $i_L = V_s/R$

Notice the pattern in Table 1 and Figure 2. In the RL circuit, $v_R = R\,i_L$ has exactly the same shape as v_C in the RC circuit, and v_L has the same shape as the RC circuit's v_R. This is why you will measure the RL time constant from v_R.

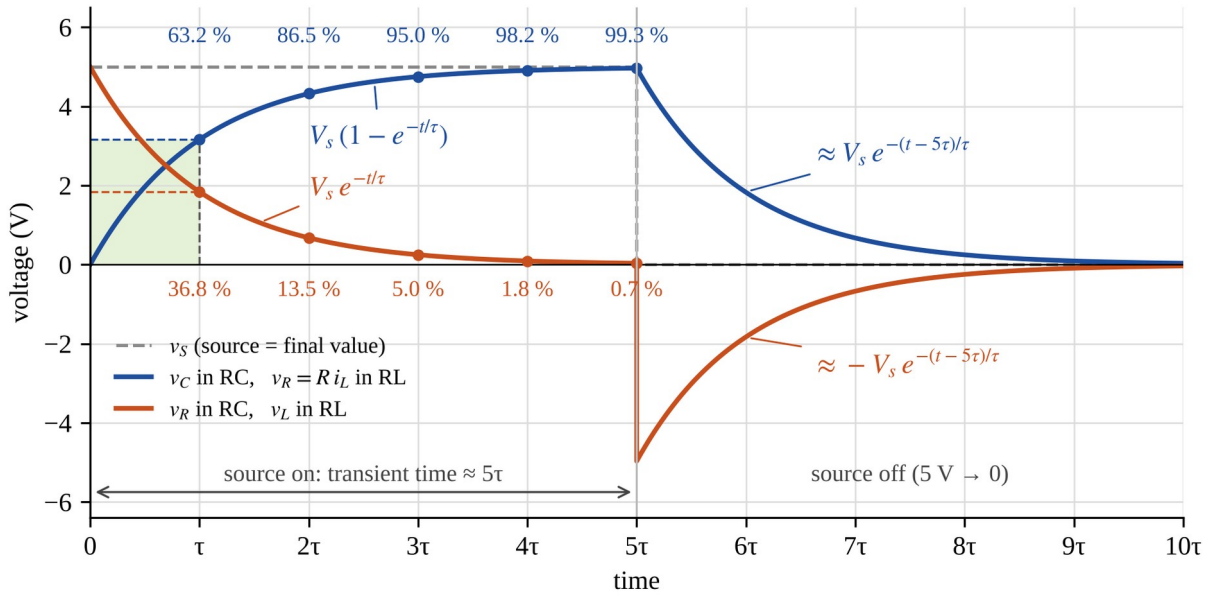


Figure 2: Step response with V_s = 5 V: the source turns on at t = 0 and off at t = 5τ. The same curves describe both circuits, with τ = RC or τ = L/R (so e^(−t/τ) = e^(−Rt/L) in the RL circuit); the inductor current is the blue curve divided by R, i_L = (V_s/R)(1 − e^(−t/τ)). Shaded: the first time constant. After t = 5τ the curves start from 99.3 % of V_s (not exactly V_s), so the decay is ≈ V_s e^(−(t−5τ)/τ), where t − 5τ is the time measured from the moment the source switches off. Blue numbers: percent of the final value reached at 1τ to 5τ. Orange numbers: percent of the initial value remaining.

Square-wave input and measuring τ

In the lab, the switch is replaced by a function generator producing a 0 to V_s square wave with period T and a 50 % duty cycle. Each edge is a new step, so the capacitor charges toward V_s while the input is high and discharges toward 0 while it is low. If each half-period is at least 5τ ($T/2 \geq 5\tau$), the capacitor charges and discharges fully every cycle. If the half-period is shorter, it never gets there; after a few cycles v_C settles into a steady pattern, rising and falling between two fixed levels V_{max} and V_{min} (Figure 3).

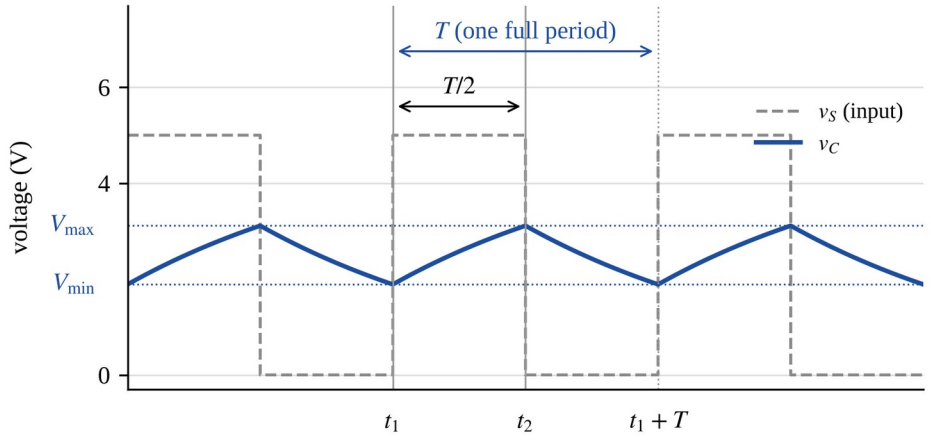


Figure 3: Steady-state capacitor voltage for a square-wave input with T = τ. From t_1 to t_2 is one half-period, T/2 = 0.5τ (much shorter than 5τ); from t_1 to t_1 + T is one full period. v_C rises and falls between the same two levels, V_{max} and V_{min}.

T and τ are different. T is the period of the square wave you set on the generator; τ is the time constant of the circuit. In the first part of each experiment you set T ≈ τ, so each half-period is only about 0.5τ, much less than the 5τ needed to fully charge. This is deliberate: the voltage never finishes charging or discharging, so it settles between two clearly separated levels V_{max} and V_{min} that you can read accurately. In step 3 you lower the frequency so each half-period is about 5τ, and you will see full charging and discharging.

During a falling half-cycle the capacitor discharges from V_{max} toward 0 for a time T/2 and arrives at V_{min}. From Table 1, $V_{min} = V_{max} e^{-(T/2)/\tau}$. Solving for τ:

$$\tau = \frac{T/2}{\ln\left(\frac{V_{max}}{V_{min}}\right)} \qquad (2)$$

Equation (2) is how you will find τ from the oscilloscope. It works well when T/2 is between about 0.5τ and 2τ. If T/2 is much longer, V_{min} is close to 0 V and hard to read accurately; if it is much shorter, V_{max} and V_{min} are almost equal. A useful check on your RC readings: for a 0 to V_s input, $V_{max} + V_{min} \approx V_s$. For the RL circuit, use the same equation on the resistor voltage v_R (see the Appendix for the derivation).

Real components: why your τ will not match exactly

Function generator: the generator has an internal output resistance of about 50 Ω in series with

your circuit. With R = 10 kΩ this adds only 0.5 %, but with R = 470 Ω it adds about 10 %. You will also see the generator's output droop slightly while the circuit draws current.

Inductor: an inductor is a long coil of wire, so it also has winding resistance R_{coil} (often tens of ohms for 100 mH). It is in series with L, so at steady state v_L settles at $i_L R_{coil}$ instead of 0, and the time constant becomes

$$\tau = \frac{L}{R + R_{coil} + R_{gen}} \tag{3}$$

Tolerance: capacitors and inductors are often only within ±5 % to ±20 % of their marked value, so measure R and R_{coil} with the DMM and use the measured values in your calculations.

Preparation

- Carefully read the Background section for this lab.
- Calculate the expected time constant for both circuits (RC: R = 10 kΩ, C = 100 nF; RL: R = 470 Ω, L = 100 mH) and the generator frequency you will need for a period of T ≈ τ and T ≈ 10τ.
- **RC circuit:** using Appendix A3 with T = τ and $V_s = 5\ V$, predict the two levels V_{max} and V_{min} that v_C will move between.
- **RL circuit:** predict τ two ways: (i) from L/R alone, and (ii) from equation (3), assuming $R_{coil} \approx 70\ \Omega$ and $R_{gen} \approx 50\ \Omega$. Which one do you expect to be closer to your measurement, and why?
- Fill in the **Predicted** rows of Table 2 and bring them to the lab.

PROCEDURE

In your report, preface each set of data with a brief description of what the data represents; provide schematics of circuits used; tabulate all measurements; include sample theoretical calculations; describe observations; provide sketches/printouts of output waveforms; comment on results.

[Before you start: set both oscilloscope channels to 10:1 to match the probes, and set the function generator's load to High Z.]

Probe grounds: both oscilloscope probe ground clips and the generator's black lead must go to the **same node**. The scope grounds are tied together inside the instrument, so a probe ground

placed anywhere else shorts out part of your circuit. To look at the voltage across a component, connect the components so that component is the one tied to ground (see Figures 4 to 7).

❖ RC CIRCUIT:

Connect **R = 10 kΩ** and **C = 100 nF (0.1 µF)** in series with the function generator. Measure the actual value of R with the DMM before you connect it. Set the generator to a **0 to 5 V square wave** with a **50 %** duty cycle and a period approximately equal to the RC time constant (τ = 1 ms, so T = 1 ms and **f ≈ 1 kHz**; each half-period is 0.5 ms = 0.5τ).

[Load: High Z
Waveform: Square
Amplitude: 5 Vpp
Offset: +2.5 V (gives 0 V to 5 V; without the offset the output swings −2.5 V to +2.5 V and equation (2) will not work)
Duty cycle: 50 %
Period: ≈ τ (RC: 1 ms → 1 kHz; RL: ≈ 0.21 ms → ≈ 4.7 kHz)]

- Record the **input voltage** $v_S(t)$ and the **capacitor voltage** $v_C(t)$ on the same oscilloscope screen. Connect R first, then C, so C is the component tied to ground.

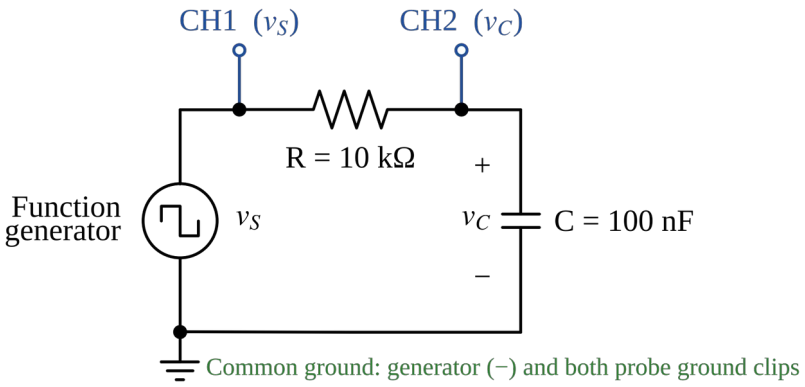


Figure 4: Measuring the capacitor voltage. C is the component tied to the common ground.

- Record the **input voltage** $v_S(t)$ and the **resistor voltage** $v_R(t)$ on the same screen. Swap the order (C first, then R) so R is the component tied to ground.

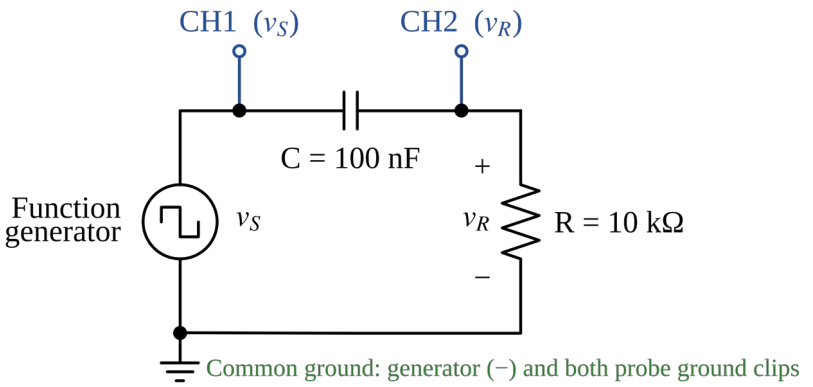


Figure 5: Measuring the resistor voltage. C and R are swapped so R is tied to the common ground.

- Compare your waveforms with Figure 8 in the Appendix to check that your set-up is correct.

How to measure V_{max}, V_{min} and T/2 with the oscilloscope cursors

Measure these on the oscilloscope, not with the DMM. At 1 kHz the voltage changes continuously, and a DMM can only show one averaged (DC) or RMS value. Use the DMM only for R and R_{coil}, with the component disconnected from the circuit.

(a) Set both channels to DC coupling and 10:1 probe attenuation. (AC coupling shifts the waveform up or down and gives wrong levels.)

(b) Trigger on CH1 (the source) so the display is stable. Adjust volts/div and position so the waveform you are measuring fills most of the screen height; the bigger it is, the more accurately you can read it.

(c) Press Cursors and set Mode = Manual and Source = the channel you are measuring (v_C in the RC circuit, v_R in the RL circuit).

(d) Y cursors: place Y1 on the lowest point of the waveform (V_{min}) and Y2 on the highest point (V_{max}).

(e) X cursors: place X1 on a rising edge of the input (t_1) and X2 on the next falling edge (t_2). The ΔX reading is T/2.

(f) Record V_{max}, V_{min} and T/2, and save a screen capture that shows the cursor readings for your report.

1. Using the cursors on your $v_C(t)$ waveform, measure V_{max}, V_{min} and $T/2 = t_2 - t_1$ (Figure 3). Calculate τ using equation (2), $\tau = \dfrac{T/2}{\ln\left(\dfrac{V_{max}}{V_{min}}\right)}$.

2. How do your measured τ and the two levels V_{max} and V_{min} compare with your predictions in

Table 2? Give the percent difference for τ, and explain any difference.

3. **Reduce** the input frequency by a factor of **10** (f ≈ 100 Hz, so T/2 ≈ 5τ) and record $v_C(t)$ and $v_R(t$ again. Compare them with the higher-frequency waveforms and explain how the lower frequency affects charging and discharging.

❖ **RL Circuit**

Connect **R = 470 Ω** and **L = 100 mH** in series with the function generator. Before connecting, measure R and the inductor's winding resistance R_{coil} with the DMM. Use the same 0 to 5 V square-wave settings (with the +2.5 V offset) and a period approximately equal to L/R (τ ≈ 213 µs, so T ≈ 213 µs and **f ≈ 4.7 kHz**; each half-period is about 107 µs ≈ 0.5τ).

- Record the **input voltage** and the **inductor voltage** $v_L(t)$ on the same screen, with L tied to ground.

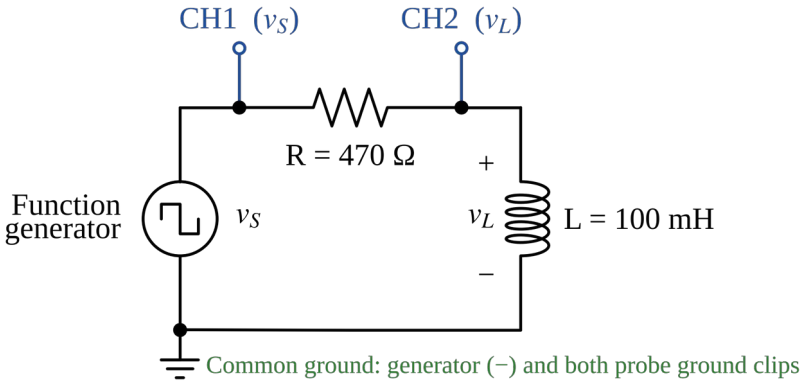


Figure 6: Measuring the inductor voltage. L is the component tied to the common ground.

- Record the **input voltage** and the **resistor voltage** $v_R(t)$ on the same screen, with R tied to ground.

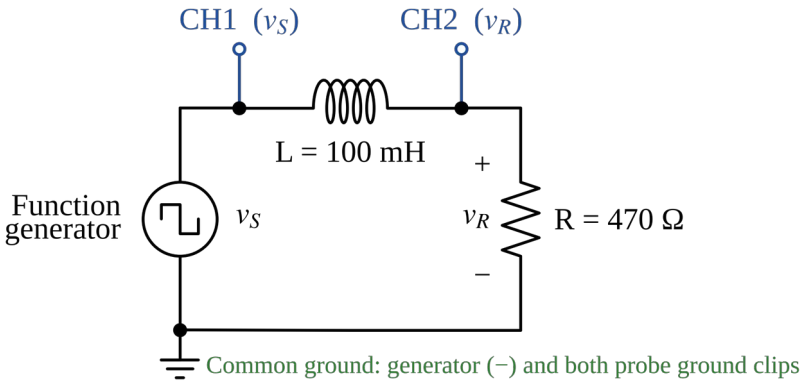


Figure 7: Measuring the resistor voltage. L and R are swapped so R is tied to the common ground.

1. Compare your waveforms with Figure 8 in the Appendix. Does $v_L(t)$ settle at exactly 0 V? Explain why or why not.
2. Measure V_{max}, V_{min} and T/2 on the **resistor voltage** $v_R(t)$, using the same cursor method as for the RC circuit, and calculate τ with equation (2). Compare it with L/R and with equation (3), $L/\left(R+R_{coil}+R_{gen}\right)$, using your measured R and R_{coil} and $R_{gen}\approx 50\ \Omega$. Which prediction is closer?
3. **Reduce** the frequency by a factor of **10** (f ≈ 470 Hz) and record $v_L(t)$ and $v_R(t)$. Compare these with the higher-frequency results.

Fill in Table 2: the **Predicted** rows before the lab (Preparation) and the **Measured** rows during the lab. % diff. compares your measured τ with your predicted τ.

Table 2: Predicted and measured results

Circuit		R	R_{coil}	τ	V_{max}	V_{min}	T/2	% diff.
RC	*Predicted*		—					—
	Measured		—					
RL	*Predicted*				—	—		—
	Measured							

For the RL circuit, enter the equation (3) value as the predicted τ; show your L/R value in your Preparation.

Discussion & Conclusion

a. Were the objectives met? Use your data to show that the lab objectives were met.
b. What else did you learn?
c. What problems did you encounter and how did you solve them?

[Submit the completed lab report. Attach *Pictures* of your work with all connections as part of your lab report]

APPENDIX: Derivations (A3 is needed for the Preparation; the rest is optional reading)

A1. Series RC circuit

KVL around the loop in Figure 1 with $i_R = i_C = C\dfrac{dv_C}{dt}$ gives

$$V_s = RC\frac{dv_C}{dt} + v_C$$

Separate the variables and integrate from $v_C(0) = V_0$ at t = 0:

$$\ln\frac{v_C(t) - V_s}{V_0 - V_s} = -\frac{t}{RC} \quad \Rightarrow \quad v_C(t) = V_s + (V_0 - V_s)e^{-t/RC}$$

This is equation (1) with $x(0) = V_0$, $x(\infty) = V_s$ and τ = RC. Setting $V_0 = 0$ gives the charging result in Table 1; setting $V_s = 0$ gives the discharging result.

A2. Series RL circuit

KVL with $v_L = L\dfrac{di_L}{dt}$ gives

$$V_s = L\frac{di_L}{dt} + R\,i_L \quad \Rightarrow \quad i_L(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-Rt/L}$$

The steps are the same as A1, and the time constant is τ = L/R. With winding and generator resistance included, R becomes $R + R_{coil} + R_{gen}$, which gives equation (3).

A3. Square-wave steady state

Let $k = e^{-(T/2)/\tau}$. In steady state, the rising half-cycle starts at V_{min} and ends at V_{max}, and the falling half-cycle starts at V_{max} and ends at V_{min}. Applying equation (1) to each half:

$$V_{max} = V_s + (V_{min} - V_s)k \qquad V_{min} = V_{max}k$$

Solving these together:

$$V_{max} = \frac{V_s}{k+1}, \qquad V_{min} = \frac{kV_s}{k+1}, \qquad \frac{V_{min}}{V_{max}} = k$$

Taking the natural log of $V_{min}/V_{max} = k$ gives equation (2), and adding the first two results shows $V_{max} + V_{min} = V_s$. As an example, if T = 2τ (each half-period is τ), k = $e^{-1} = 0.368$, so with $V_s = 5\ V$ you would expect $V_{max} \approx 3.66\ V$ and $V_{min} \approx 1.34\ V$.

A4. What you should see on the oscilloscope

Figure 8 shows the ideal waveforms for both settings you will use, with a 0 to 5 V square wave. At T ≈ τ the half-period is too short for the circuit to settle, so v_C (and v_R in the RL circuit) moves between V_{max} and V_{min}. At T ≈ 10τ each half-period is about 5τ, so the circuit settles almost completely (over 99 %) before every edge. The RL plots are identical to the RC plots, with v_R in place of v_C and v_L in place of v_R.

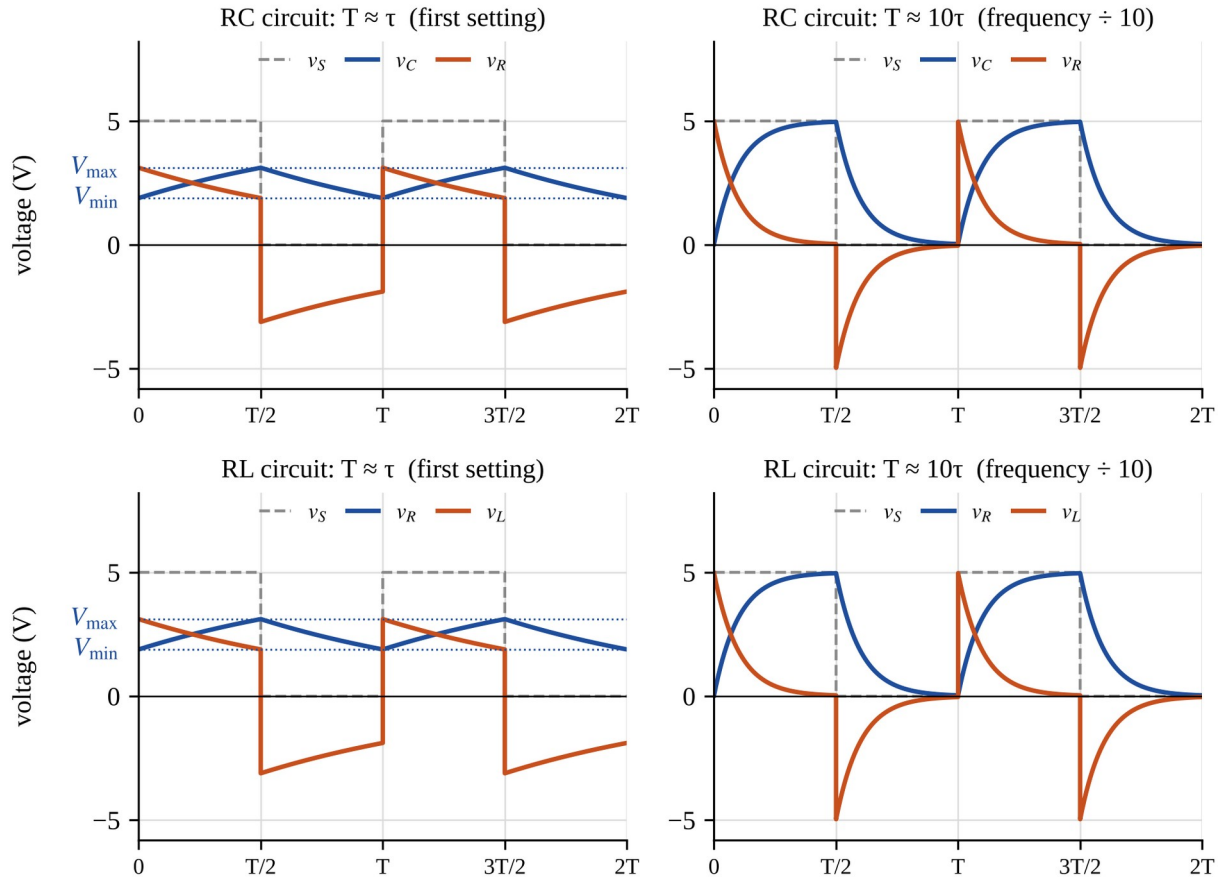


Figure 8: Expected waveforms (ideal components). Top: RC circuit. Bottom: RL circuit. Left: T ≈ τ. Right: T ≈ 10τ.

Your real waveforms will differ slightly. In the RL circuit, v_L settles at a small positive voltage ($i_L R_{coil}$) instead of 0 V, and v_R never quite reaches 5 V. The generator output also droops a little while current flows. Equation (3) and the “Real components” section explain why.